

SCALLING THE ANALYSIS: THE EXPANDED COHORT

4.1 Transition to the Larger Dataset

All supervised and unsupervised experiments described in the preceding chapters were developed, tuned, and evaluated on the *pilot cohort*: the 244-sample, 462-feature dataset assembled in Chapter 3 from a TCGA-BRCA data release that had been pre-processed and curated by our partners at Nova Medical School prior to the start of this project. This was the only dataset available at that stage of the work. At a later point in the project, access was obtained to a substantially larger and independently sourced dataset from the same TCGA-BRCA study, retrieved directly from the cBioPortal for Cancer Genomics. The availability of this expanded cohort created a natural opportunity to revisit the best-performing pipelines developed on the pilot data and assess whether the findings generalise to a larger and more representative sample of the same disease population. This section documents the origin, composition, and relationship to the pilot cohort of the expanded dataset that forms the basis of the analysis presented in the remainder of this chapter.

Data Source and Acquisition

The expanded dataset was obtained directly from the cBioPortal for Cancer Genomics [15, 16], an open-access platform that provides curated multi-omics data from The Cancer Genome Atlas (TCGA) [61] and other large-scale consortia. Specifically, the miRNA expression and clinical data for the TCGA Breast Invasive Carcinoma (TCGA-BRCA) study were downloaded from the portal’s data export interface. Unlike the pilot cohort, which arrived as a pre-processed artefact from an established collaboration, this release was obtained at a later stage in the project and represents a direct, largely unprocessed export of the TCGA-BRCA repository. The clinical table and the miRNA expression matrix were retrieved separately and subsequently merged by patient identifier, as described in the preprocessing subsection below.

After integration, the raw combined dataset contained $N = 773$ samples described by 539 miRNA expression features and 15 clinical variables, plus the PAM50 subtype label.

Following the application of the preprocessing steps documented in Section 4.2, the final modelling cohort comprises **702 samples**, making it approximately 2.9 times larger than the pilot cohort.

Assessing the Intersection with the Pilot Cohort

Because the two datasets were obtained through different channels and exhibit differences in both nomenclature and feature composition (detailed in the next subsection), it was necessary to formally verify that they originate from the same underlying study before drawing any comparative conclusions. This verification was conducted by cross-referencing the patient identifiers present in each dataset.

TCGA assigns every biospecimen a standardised barcode of the form TCGA-{TSS}-{Participant}-{Sample}, where the first twelve characters (TCGA-XX-XXXX) uniquely identify a patient, independent of sample type or processing version. The pilot cohort uses the full barcode including the two-digit sample code suffix (*e.g.*, TCGA-A1-A0SK-01), while the expanded dataset stores only the patient-level prefix (*e.g.*, TCGA-A1-A0SK). After normalising both sets of identifiers to the twelve-character patient prefix, we can visualize in Figure 4.1 that **176 of the 244 patients in the pilot cohort (72.1%)** were found to have a direct match in the expanded dataset. The remaining 68 pilot-cohort patients are absent from the expanded release, most likely because they correspond to samples that were present in the specific TCGA data version used by Nova Medical School but are not included in the cBioPortal export retrieved at a later date, or because they were filtered out by the curation pipeline applied at the source.

A 72% overlap between an independently acquired dataset and a pre-processed cohort sourced from the same TCGA study is consistent with the expected degree of overlap between two releases of the same underlying data. The patients present in the pilot cohort but absent from the expanded release can be attributed to version differences in the TCGA data freeze or to curation decisions applied upstream, rather than to any fundamental incompatibility between the two sources.

Differences Between the Two Datasets

Despite originating from the same repository and study, the two datasets differ in several important respects. These differences are a direct consequence of the independent preprocessing applied to each release, and understanding them is necessary to interpret any comparative results.

miRNA Feature Space. The pilot cohort was built from a raw TCGA-BRCA miRNA expression matrix containing 888 features, which were reduced to 442 unique, non-redundant miRNAs after the deduplication step described in Section 3.2.2. The features follow a HUGO gene symbol convention with isoform suffixes (*e.g.*, MIR-744/5P, MIR-744/3P). The expanded dataset, by contrast, already provides 539 miRNA features in a pre-deduplicated

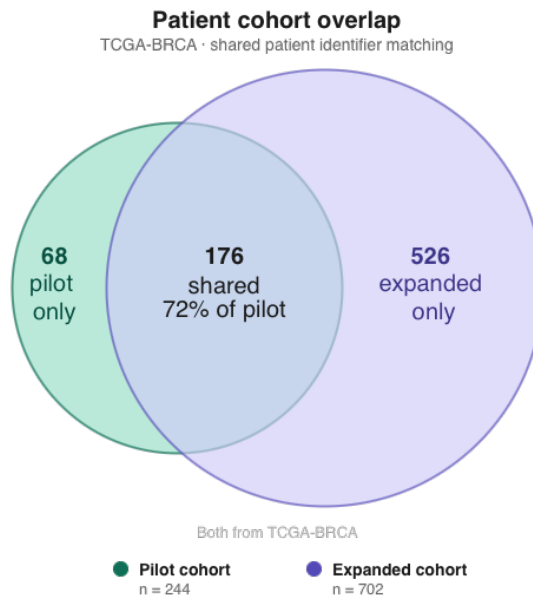


Figure 4.1: This is a description of my image.

form, using the standard `hsa-mir-xxx` Ensembl/miRBase nomenclature (e.g., `hsa-mir-744`). The difference in feature count (442 vs. 539) reflects both the use of a different TCGA data release and the different deduplication strategy applied at source: the cBioPortal export retains one entry per unique miRNA base identifier, whereas the pilot cohort was filtered by matching expression profiles across samples. Although the two naming conventions refer to the same biological entities, a direct one-to-one mapping between them is not guaranteed without an explicit identifier translation step, as the cBioPortal identifiers represent precursor miRNA genes while the pilot-cohort identifiers can encode strand-specific isoforms.

Clinical Feature Set. The clinical annotations available in the two datasets are largely complementary rather than overlapping. The pilot cohort contains a rich set of molecular and integrative features curated by Nova Medical School, including receptor status indicators (`ER_STATUS`, `PR_STATUS`, `HER2_STATUS`), TCGA multi-omics cluster assignments (`CN_CLUSTER`, `METHYLATION_CLUSTER`, `MIRNA_CLUSTER`), and detailed TNM staging variables. The expanded dataset, drawn directly from the cBioPortal clinical export, provides a different profile of 15 variables centred on genomic instability and tumour microenvironment metrics: the Aneuploidy Score, three independent hypoxia scores (Buffa, Ragnum, and Winter), MSI (Microsatellite Instability) scores from two platforms (MSI MANTIS and MSIsensor), Tumour Break Load, and TMB (non-synonymous). Both datasets share a common core of basic clinical descriptors: patient age at diagnosis, tumour stage, lymph node involvement, distant metastasis, fraction of the genome altered, mutation count, and TMB; which provides a minimal comparable clinical basis across experiments. The key differences in feature availability between the two datasets are summarised in Table 4.1.

Table 4.1: Comparison of the key characteristics of the pilot cohort (Chapter 3) and the expanded cohort used in this chapter. Dashes indicate that the feature category is not available in that dataset.

Property	Pilot Cohort	Expanded Cohort
Scale		
Samples	244	702
miRNA features	442	539
Clinical features	20	15
Total features	462	554
Subtype distribution		
Luminal A	103 (42.2%)	362 (51.6%)
Luminal B	71 (29.1%)	156 (22.2%)
Basal-like	41 (16.8%)	117 (16.7%)
HER2-enriched	29 (11.9%)	67 (9.5%)
Clinical feature profile		
Receptor status (ER/PR/HER2)	✓	—
Multi-omics cluster assignments	✓	—
Detailed TNM staging variables	✓	✓
Hypoxia scores (Buffa/Ragnum/Winter)	—	✓
MSI scores (MANTIS / MSIsensor)	—	✓
Aneuploidy Score	—	✓
Tumour Break Load	—	✓
Overall survival	✓	—
miRNA nomenclature		
Format	MIR-xxx/isoform	hsa-mir-xxx
Patient overlap		
Shared patients (12-char prefix)	176 (72% of pilot cohort)	
Patient demographics		
Age range (years)	26–90	26–90
Mean ± Std (years)	59.3 ± 13.3	58.1 ± 13.4
Median age (years)	60	58

Characteristics of the Expanded Dataset

The final modelling dataset contains 702 samples described by 554 features: 539 miRNA expression values and 15 clinical variables. All features are fully numeric and there are no missing values, as the preprocessing steps described below removed samples or imputed values where necessary prior to modelling.

Subtype Distribution. As shown in Table 4.1 and Figure 4.2, the class distribution follows the expected epidemiological pattern, with Luminal A comprising the largest fraction of the cohort ($n = 362$; 51.6%), followed by Luminal B ($n = 156$; 22.2%), Basal-like ($n = 117$; 16.7%), and HER2-enriched ($n = 67$; 9.5%). Compared with the pilot cohort, the expanded dataset shows a more pronounced Luminal A dominance (51.6% vs. 42.2%), which can partially be attributed to the absence of the Normal-like class, which had been explicitly excluded during the preprocessing of both cohorts. The Normal-like class, as seen in the smaller cohort, has a very low sample size that will unable our models and implementations to find a meaningful feature group that could represent this class. The class imbalance is less severe than in the pilot cohort in relative terms, but it remains a relevant modelling challenge: a trivial majority-class classifier would achieve 51.6% accuracy without learning any discriminative signal. The same class-imbalance mitigation strategies employed in the pilot study (stratified cross-validation and class-weighted loss functions) are therefore retained in all experiments described in this chapter.

PAM50 Subtype	Count	%
Luminal A	383	49.5
Luminal B	161	20.8
Basal-like	133	17.2
HER2-enriched	70	9.1
Normal-like	26	3.4
Total	773	100.0

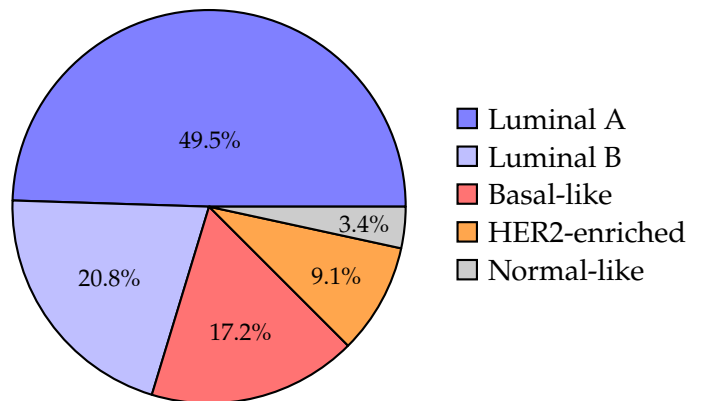


Figure 4.2: PAM50 subtype distribution across the 773-sample Bigger cohort. Luminal A remains the dominant class (almost 50%), while Normal-like continues severely under-represented ($n = 26$; 3.4%).

Clinical Feature Profile. The 15 clinical variables available in this dataset are described in Table 4.3. As noted above, this set differs substantially from the pilot cohort’s clinical annotation. While that doesn’t mean that we our classification have been more heavily reliant on miRNA expression, it does mean that we have had to rely on other clinical features to help us classify the subtypes.

Table 4.3: Clinical features available in the expanded cohort, with data type and brief clinical description.

Feature	Type	Description
Diagnosis Age	Integer	Patient age at diagnosis (years).
Neoplasm Stage (AJCC)	Ordinal	Overall AJCC composite stage at diagnosis.
Aneuploidy Score	Continuous	Number of arm-level copy-number alterations; proxy for chromosomal instability.
Buffa Hypoxia Score	Continuous	Transcriptomic hypoxia signature (Buffa <i>et al.</i>).
Ragnum Hypoxia Score	Continuous	Transcriptomic hypoxia signature (Ragnum <i>et al.</i>).
Winter Hypoxia Score	Continuous	Transcriptomic hypoxia signature (Winter <i>et al.</i>).
Fraction Genome Altered	Continuous	Proportion of genome with somatic copy-number alterations.
MSI MANTIS Score	Continuous	Microsatellite instability score (MANTIS algorithm).
MSIsensor Score	Continuous	Microsatellite instability score (MSIsensor tool).
Mutation Count	Integer	Total somatic mutation count.
Metastasis Stage (AJCC)	Binary	Presence of distant metastasis (M0/M1).
Lymph Node Stage (AJCC)	Ordinal	Regional lymph node involvement (N0–N3).
Tumour Stage (AJCC)	Ordinal	Primary tumour size/extent (T1–T4).
Tumour Break Load	Continuous	Number of structural variant breakpoints; proxy for genomic instability.
TMB (nonsynonymous)	Continuous	Tumour mutational burden (nonsynonymous mutations per Mb).

4.2 Data Pre-processing Pipeline

The expanded dataset was sourced as two separate files from cBioPortal: a miRNA expression matrix (773 patients $\times$ 539 features) and a clinical annotations table (773 patients $\times$ 21 columns). Both arrived in patient-as-rows orientation and shared a common Patient ID key, so no transposition was needed before merging (unlike the pilot cohort), where the miRNA matrix required structural realignment (Section 3.2).

Column removal. Four clinical columns were removed prior to joining: Ethnicity Category (14.3% missing), Genetic Ancestry Label (12.2% missing), and Race Category (6.5% missing) due to substantial missingness, and Cancer Type Detailed as a task-irrelevant

field (all samples are TCGA-BRCA primaries). This left 15 clinical variables for modelling.

Integration, subtype cleaning, and missing value handling. The two files were merged via an inner join on Patient ID. Since both files contained the same 773 patients, the join introduced no sample loss. The 26 Normal-like samples (BRCA_Normal) were then excluded, leaving 747 samples. Residual missing values remained in three clinical features after this exclusion: MSI MANTIS Score ($n = 24$), Aneuploidy Score ($n = 17$), and Neoplasm Disease Stage ($n = 4$). As in the pilot cohort, listwise deletion was applied rather than imputation, given the clinical nature of these variables. This removed a further 45 samples, yielding a final cohort of **702 samples**.

Categorical encoding and miRNA renaming. The remaining categorical clinical columns (TNM staging codes, metastasis stage, and lymph node stage) were label-encoded using sklearn's LabelEncoder. The PAM50 subtype was mapped to integers consistent with the pilot cohort (Basal-like $\rightarrow 0$, HER2-enriched $\rightarrow 1$, Luminal A $\rightarrow 2$, Luminal B $\rightarrow 3$). miRNA column identifiers were renamed by stripping the hsa- prefix and capitalising mir to MIR (e.g., hsa-mir-577 $\rightarrow$ MIR-577) for readability.

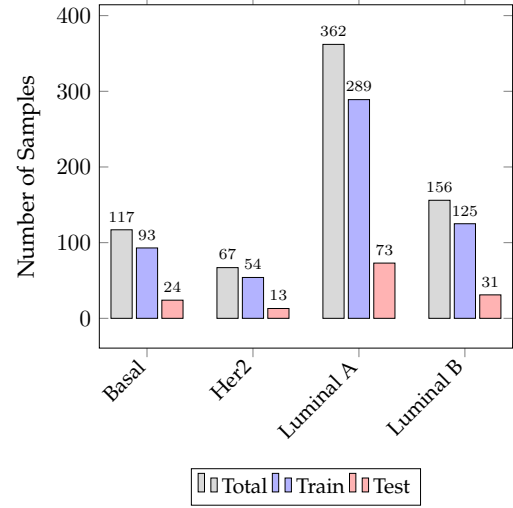
Feature matrix and no-scaling decision. The final feature matrix $\mathbf{X}$ comprises 554 features (539 miRNA + 15 clinical) and the target vector $\mathbf{y}$ the integer-encoded subtype. The StandardScaler was applied on the continuous features: all 539 miRNA expression columns and the continuous clinical variables, while the integer-encoded ordinal and binary clinical columns were left unscaled as their values are already bounded and comparable in magnitude. Consistent with the data leakage prevention protocol described in Section 3.2.6, the scaler was fitted exclusively on the training fold and subsequently applied without refitting to both the training and test folds.

Train/test split. The dataset was partitioned into an 80/20 stratified split, yielding 561 training and 141 test samples, with subtype proportions preserved across both partitions, as shown in Table 4.3a.

Comparison with the pilot cohort pipeline. Table 4.4 summarises the steps applied to each cohort. The key omissions in the expanded pipeline are miRNA deduplication (the cBioPortal export already provides one entry per unique miRNA) and feature standardisation (unnecessary for tree-based models). The key addition is the WITH_CLINICAL flag, which makes the clinical/miRNA-only comparison explicit.

Property	Value
<i>Dimensionality</i>	
Total samples	702
miRNA features	539
Clinical features	15
Total features	554
Feature-to-sample ratio (p/n)	0.79
<i>Data quality</i>	
Missing values	0
miRNAs with variance > 1.0	147 (27.3%)
<i>miRNA expression ($\log_2(x+1)$ scale)</i>	
Range	[0.00, 19.37]
Mean $\pm$ std	3.11 $\pm$ 3.88
Median	1.28
<i>Clinical continuous features</i>	
Age (years)	58.1 $\pm$ 13.4 [26, 90]
Mutation count	79.4 $\pm$ 215.9 [3, 3879]
Fraction genome altered	0.31 $\pm$ 0.21 [0.00, 1.00]
TMB non-synonymous	2.76 $\pm$ 7.29 [0.10, 130.30]
Aneuploidy score	12.5 $\pm$ 7.7 [0, 34]

(a) Summary statistics of the final dataset.



(b) Subtype distribution histogram.

Figure 4.3: Overview of the final modelling dataset for the expanded cohort: (a) detailed summary of clinical and molecular features; (b) distribution of PAM50 subtypes across the full dataset and the stratified train/test partitions.

Table 4.4: Preprocessing steps applied to the pilot cohort (Chapter 3) and the expanded cohort. A dash indicates the step was not required.

Step	Pilot cohort	Expanded cohort
miRNA matrix transposition	✓	—
Inner join (miRNA + clinical)	✓	✓
miRNA deduplication	✓	—
Removal of uninformative columns	✓	✓
Normal-like exclusion	✓	✓
Listwise deletion	✓	✓
Categorical encoding	✓	✓
Subtype integer encoding	✓	✓
miRNA column renaming	—	✓
With-/without-clinical flag	—	✓
Stratified 80/20 split	✓	✓
Feature standardisation	✓	—

4.3 Reassessing the Methodology at Scale

The preceding sections of this chapter have established the foundations necessary to carry the analysis forward. Section 4.2 documented the origin, composition, and preprocessing of the expanded cohort: a 702-sample, 554-feature dataset drawn independently from the TCGA-BRCA repository, encompassing 539 miRNA expression features and 15 clinical variables of a qualitatively different nature from those available in the pilot study. The dataset is substantially larger, differently annotated, and independently assembled and yet, it originates from the same patient population where 72% of the patients in the pilot cohort share the same identifier as the ones on this study.

With the data prepared and characterised, the natural question that follows is whether the patterns identified in Chapter 3 hold at this larger scale. The pilot study produced two sets of findings that are worth reassessing. The first is methodological: XGBoost under a One-vs-Rest formulation, validated against ANOVA F-test feature rankings, proved effective both as a classifier and as a biomarker discovery proxy. The second is biological: a consistent separability gradient emerged across the four PAM50 subtypes, with Basal-like being the most reliably identified and Luminal B the least, and with a small set of miRNAs - MIR-577 prominently among them - accounting for a disproportionate share of the model's discriminative gain for the most separable subtype. The unsupervised K-Means analysis independently corroborated this picture, isolating Basal-like into a high-purity cluster without any label supervision.

This section reapplies those same implementations - the XGBoost pipeline, the ANOVA F-test sweep, and the K-Means clustering - to the expanded cohort without modification to their architecture or parameterisation. Holding the methodology fixed is a deliberate choice: the goal here is not to optimise performance on the new data, but to test **generalisation**. If the separability gradient, the dominant feature panels, and the unsupervised cluster structure are genuine properties of Basal-like and the other PAM50 subtypes, they should be recoverable from an independently assembled dataset of greater size. If they are not, the pilot findings would need to be qualified as cohort-specific. The results of this reassessment form the empirical basis for the cross-cohort comparison and discussion presented in Chapter 5.

4.3.1 Per-Subtype Classification Performance

The comparison tables presented above give a global picture of how the pipeline behaves on the expanded cohort, but aggregate metrics can mask very different stories playing out at the subtype level. Recall that the central motivation for using a One-vs-Rest formulation is precisely to expose this variation: each binary classifier is trained to distinguish one subtype from all others, and its performance directly reflects how clearly that subtype's miRNA expression signature separates it from the rest of the cohort. Understanding *which subtypes are easy to identify and which are not* is not merely a performance observation, it

is a biological claim about the geometry of the miRNA expression space, and it is the foundation on which the feature extraction analysis in Section 4.4 rests. A subtype that is reliably classified is also a subtype whose most important features are meaningful; a subtype that is not can only yield tentative biomarker candidates.

The following paragraphs examine each subtype in turn, connecting the pilot-cohort findings from Section 3.3.6 to the expanded-cohort results, so that the reader can assess what held, what improved, and what remained a persistent challenge.

Basal-Like In the pilot study, the Basal-like OvR classifier achieved perfect classification under both feature configurations: precision, recall, and F1-score of 1.00 across the board, which was attributed to the subtype’s well-known and distinct triple-negative miRNA signature. The natural question when moving to the expanded cohort is whether this perfection was a function of the small pilot test set (only 8 Basal-like samples in a 49-sample test split) or a genuine reflection of the subtype’s separability.

On the 141-sample expanded test set (with 24 Basal-like cases) the answer is unambiguous. In miRNA-only configuration, the classifier achieves precision 1.00, recall 0.96, and F1-score 0.98; with clinical features added, it recovers to a perfect F1-score of 1.00. The marginal drop in recall observed without clinical data (four missed Basal-like samples out of 24) vanishes entirely when clinical covariates are included. This performance, maintained across a nearly three-fold larger test set, confirms that the pilot result was not a small-sample artifact: Basal-like tumours occupy a geometrically isolated region of the miRNA expression space, and the OvR classifier reliably finds it. As will be detailed in Section 4.4.1, MIR-577 remains the dominant discriminative feature for this subtype across both conditions and both cohorts.

Table 4.5: Basal-like OvR classifier performance across cohorts and feature configurations.

Cohort	Configuration	Precision	Recall	F1-score
Pilot ($n_{\text{test}} = 49$, 8 Basal)	miRNA only	1.00	1.00	1.00
Pilot ($n_{\text{test}} = 49$, 8 Basal)	miRNA + clinical	1.00	1.00	1.00
Expanded ($n_{\text{test}} = 141$, 24 Basal)	miRNA only	1.00	0.96	0.98
Expanded ($n_{\text{test}} = 141$, 24 Basal)	miRNA + clinical	1.00	1.00	1.00

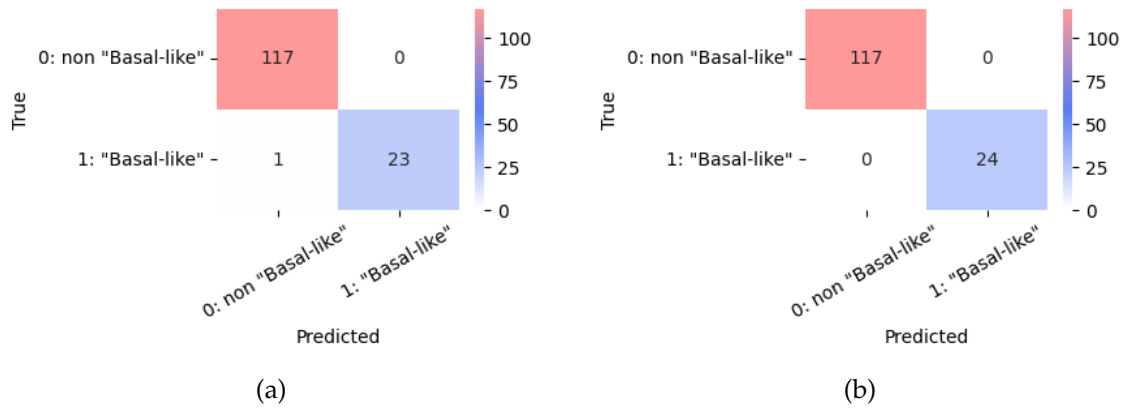


Figure 4.4: Comparison of Classification Performance for Basal-like Breast Cancer Sub-type classification where (a) has only miRNA data and (b) has both miRNA and clinical data.

4.3.1.1 HER2-Enriched

The pilot study identified HER2-enriched as the most challenging subtype, with the clinical improvement observed at that stage largely driven by HER2_STATUS acting as a direct label proxy rather than by any richer biological signal (Section 3.3.8). The expanded cohort removed that shortcut, as HER2_STATUS was not available in the cBioPortal clinical export (Section 4.2), making these results a more honest test of what miRNAs expression and generic genomic features could achieve without receptor status. Table 4.6 reports the performance obtained across both cohorts and configurations. On the expanded test set, the miRNA-only performance was consistent with the pilot, while the addition of clinical covariates yielded a gain of +0.15 in F1-score for the positive class. This improvement was driven by genomic instability metrics rather than receptor status: the Buffa Hypoxia Score emerged as the second most important feature for HER2-enriched classification in the combined condition (Section 4.4.1), suggesting a biologically plausible association between HER2 amplification, angiogenic dysregulation, and tumour hypoxia. Despite this, HER2-enriched remained the clearest bottleneck in the pipeline, with its consistent difficulty across both cohorts, at two different scales and with and without receptor status, pointing to a fundamental property of its transcriptomic landscape rather than to any correctable data limitation.

Table 4.6: HER2-enriched OvR classifier performance across cohorts and feature configurations.

Cohort	Configuration	Precision	Recall	F1-score
Pilot ($n_{\text{test}} = 49$, 6 HER2)	miRNA only	1.00	0.33	0.50
Pilot ($n_{\text{test}} = 49$, 6 HER2)	miRNA + clinical	1.00	0.33	0.50
Expanded ($n_{\text{test}} = 141$, 13 HER2)	miRNA only	0.67	0.31	0.42
Expanded ($n_{\text{test}} = 141$, 13 HER2)	miRNA + clinical	0.75	0.46	0.57

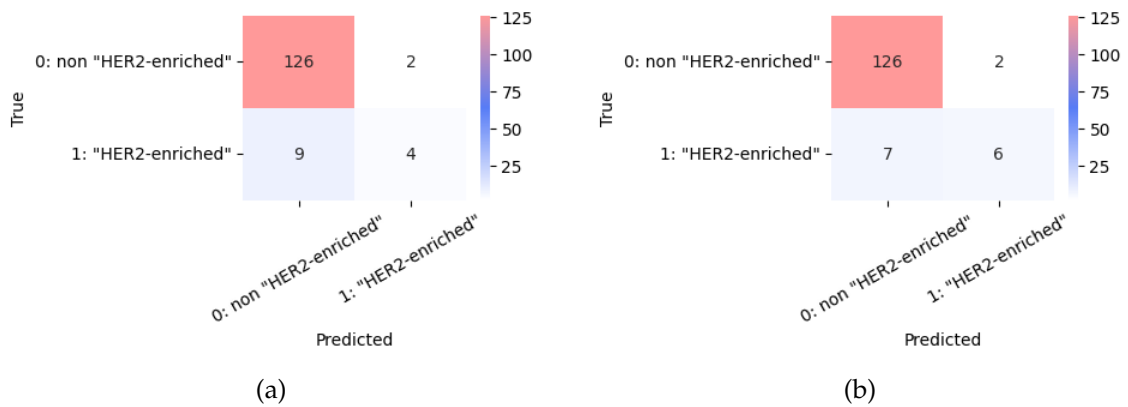


Figure 4.5: Comparison of Classification Performance for HER2-enriched Breast Cancer Subtype classification where (a) has only miRNA data and (b) has both miRNA and clinical data.

4.3.1.2 Luminal A

Luminal A was reliably classified in the pilot study, with macro-averaged F1-scores consistent with its biological distinctiveness as the low-proliferation, hormone receptor-positive subtype. When scaling to the expanded cohort, the primary concern was class dominance: with Luminal A comprising 51.6% of the cohort (compared to 42.2% in the pilot), high recall could plausibly reflect a majority-class bias rather than genuine learning. Table 4.7 shows that this was not the case. Performance on the expanded cohort mirrored the pilot almost exactly across both configurations, and the high recall obtained under the miRNA-only condition ruled out a majority-class bias explanation, as it required the model to genuinely learn the class signature. The near-equivalence of the two feature configurations suggested that the available clinical features carried little additional signal for this subtype, contrasting with the pilot study where receptor status variables provided a measurable improvement. Despite this stable classification performance, Luminal A remained without a well-defined, compact miRNA expression signature: as observed in the unsupervised analysis, it fragmented into two distinct clusters under both conditions, and the discriminative gain was spread across multiple features rather than concentrated in a small dominant panel.

Table 4.7: Luminal A OvR classifier performance across cohorts and feature configurations.

Cohort	Configuration	Precision	Recall	F1-score
Pilot ($n_{\text{test}} = 49$, 21 Luminal A)	miRNA only	0.81	0.81	0.81
Pilot ($n_{\text{test}} = 49$, 21 Luminal A)	miRNA + clinical	0.75	0.86	0.80
Expanded ($n_{\text{test}} = 141$, 73 Luminal A)	miRNA only	0.67	0.31	0.42
Expanded ($n_{\text{test}} = 141$, 73 Luminal A)	miRNA + clinical	0.75	0.46	0.57

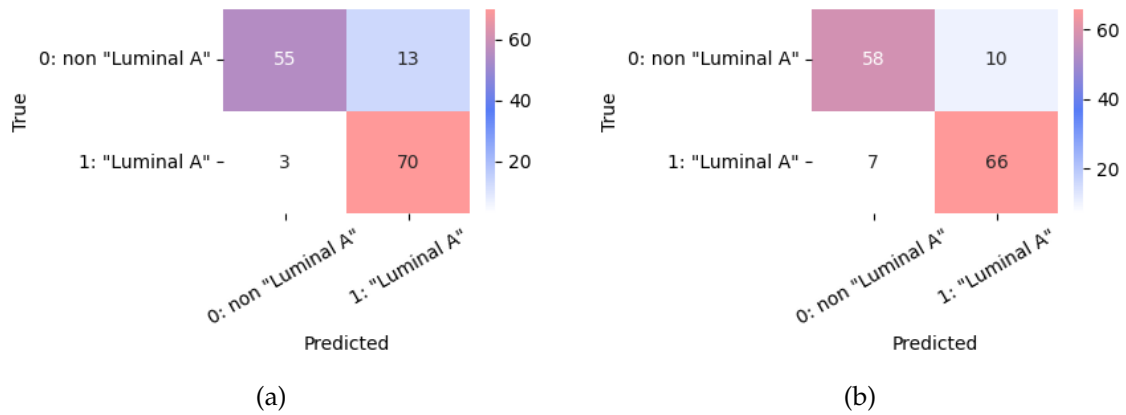


Figure 4.6: Comparison of Classification Performance for Luminal A Breast Cancer Sub-type classification where (a) has only miRNA data and (b) has both miRNA and clinical data.

4.3.1.3 Luminal B

In the pilot study, Luminal B was the second hardest subtype, with its difficulty attributed to its transcriptomic proximity to Luminal A: the two subtypes share the hormone receptor-positive programme and differ mainly in proliferative index, a difference not cleanly encoded in miRNA expression alone. The clinical improvement observed at that stage was driven primarily by ER_STATUS, which partially distinguished Luminal B from Luminal A through their different hormonal profiles. Table 4.8 shows that the expanded cohort reproduced this picture faithfully. The miRNA-only F1-score dropped from 0.50 to 0.42 on the expanded test set, with a corresponding decrease in both precision ($0.83 \rightarrow 0.67$) and recall ($0.36 \rightarrow 0.31$), consistent with the increased difficulty of classifying a larger and more heterogeneous Luminal B population without receptor status information. The addition of clinical features yielded a gain of +0.15 in F1-score for the positive class in the expanded cohort, achieved without receptor status, with the Ragnum Hypoxia Score and MSI-related features providing a modest but consistent supplementary signal (Section 4.4.1). The fact that Luminal B remained the hardest subtype across both cohorts and both configurations confirms that its difficulty is not a statistical power problem that more

data could resolve, but a reflection of the subtype’s genuine transcriptomic overlap with Luminal A. Taken together, the four subtypes continued to follow the same separability gradient identified in the pilot study: **Basal-like** > **Luminal A** > **Luminal B** $\approx$ **HER2-enriched**, a pattern that held across two independent datasets of different sizes, annotations, and preprocessing histories.

Table 4.8: Luminal B OvR classifier performance across cohorts and feature configurations.

Cohort	Configuration	Precision	Recall	F1-score
Pilot ($n_{\text{test}} = 49$, 14 Luminal B)	miRNA only	0.83	0.36	0.50
Pilot ($n_{\text{test}} = 49$, 14 Luminal B)	miRNA + clinical	0.88	0.50	0.64
Expanded ($n_{\text{test}} = 141$, 31 Luminal B)	miRNA only	0.67	0.31	0.42
Expanded ($n_{\text{test}} = 141$, 31 Luminal B)	miRNA + clinical	0.75	0.46	0.57

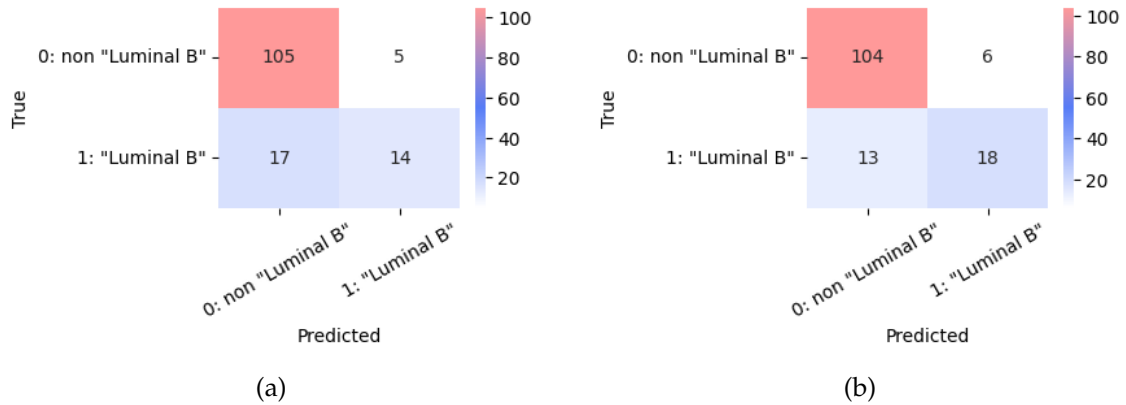


Figure 4.7: Comparison of Classification Performance for Luminal B Breast Cancer Subtype classification where (a) has only miRNA data and (b) has both miRNA and clinical data.

4.4 Feature Extraction and Statistical Corroboration

The supervised classification results in the previous section establish that the XGBoost pipeline generalises well to the expanded cohort, at least for subtypes with strong transcriptomic signatures. However, strong classification performance is only a means to an end: the model is used here as a **proxy** whose internal structure can be interrogated to retrieve the miRNAs and clinical variables that carry the most subtype-specific discriminative information. Accordingly, this section applies the same two-level feature extraction and validation strategy developed in Section 3.3.7 and Section 3.3.8: per-subtype gain importances from the XGBoost booster, followed by an independent ANOVA F-test to assess whether the highest-ranked features are also **individually** and **statistically** separable across subtypes.

As in the pilot study, both methods are treated as complementary rather than redundant. XGBoost gain captures the incremental predictive value of a feature *within the model's multivariate decision process*; ANOVA F-test captures the marginal statistical separation of a feature across subtype groups *in isolation*. A feature that is selected by both methods independently constitutes a stronger biomarker candidate than one flagged by either method alone.

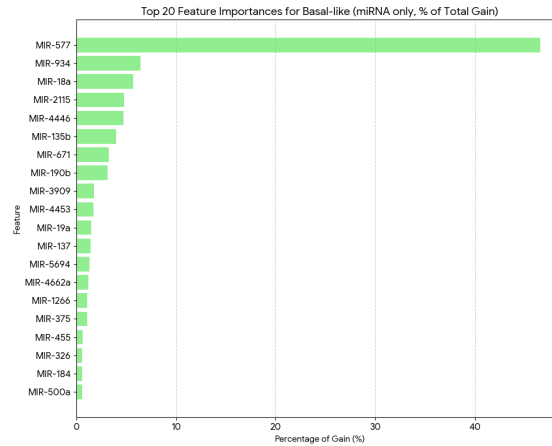
4.4.1 XGBoost Feature Importances

Following the procedure described in Section 3.3.7, the internal XGBoost booster was used to extract per-subtype gain importances for each OvR binary classifier. The top 20 features by total gain are reported below for each subtype and each feature configuration.

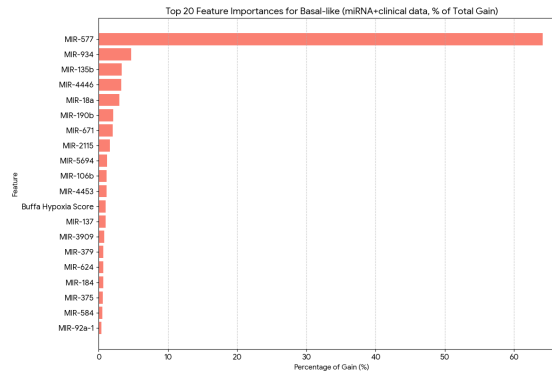
4.4.1.1 Basal-Like Feature Panel

The Basal-like gain importance landscape in the expanded cohort is dominated by **MIR-577** to an even greater degree than in the pilot study. In the miRNA-only configuration, MIR-577 accounts for 46.58% of the total gain, already a substantial majority; with clinical features added, its share rises further to 64.15%. This dominance is consistent across both cohorts and both feature configurations, making MIR-577 the **most robustly supported candidate discriminative miRNA** for the Basal-like subtype in this work.

Beyond MIR-577, the core panel is stable: MIR-934 (6.45%), MIR-18a (5.68%), MIR-2115 (4.76%), MIR-4446 (4.71%), MIR-135b (4.00%), MIR-671 (3.27%), and MIR-190b (3.11%) complete the top eight in the miRNA-only condition, and are all replicated in the miRNA + clinical ranking. The collective gain of the top-20 features reaches 92.0% in the miRNA-only condition and 94.0% with clinical data, indicating an unusually concentrated importance structure consistent with a subtype-specific miRNA programme rather than diffuse expression differences.



(a) miRNA-only feature configuration



(b) miRNA + clinical features configuration

Figure 4.8: Top 20 XGBoost gain importances for the Basal-like subtype in the expanded cohort, shown side by side for both feature configurations.

4.4.1.2 HER2-Enriched Feature Panel

The HER2-enriched importance landscape is considerably more distributed, with no single feature dominating. In the miRNA-only configuration, the leading feature MIR-4728 captures only 8.17% of total gain, and the top-20 features together account for 57.1%. With clinical features, the importance becomes slightly more concentrated (top-20 gain: 64.7%), with the Buffa Hypoxia Score entering at second place (6.62%), following MIR-4791 (8.38%). The emergence of a transcriptomic hypoxia signature as a discriminative clinical feature for HER2-enriched tumours is biologically plausible, given the known association between HER2 amplification, angiogenic dysregulation, and tumour hypoxia. Other consistent top-ranking miRNAs across both configurations include MIR-2115, MIR-6733, MIR-2355, and MIR-4728.

The distributed importance structure for HER2-enriched directly mirrors the subtype's classification difficulty: a subtype that cannot be cleanly separated from a small set of dominant features will naturally produce a classifier with limited recall, as observed in Section 4.3.1.

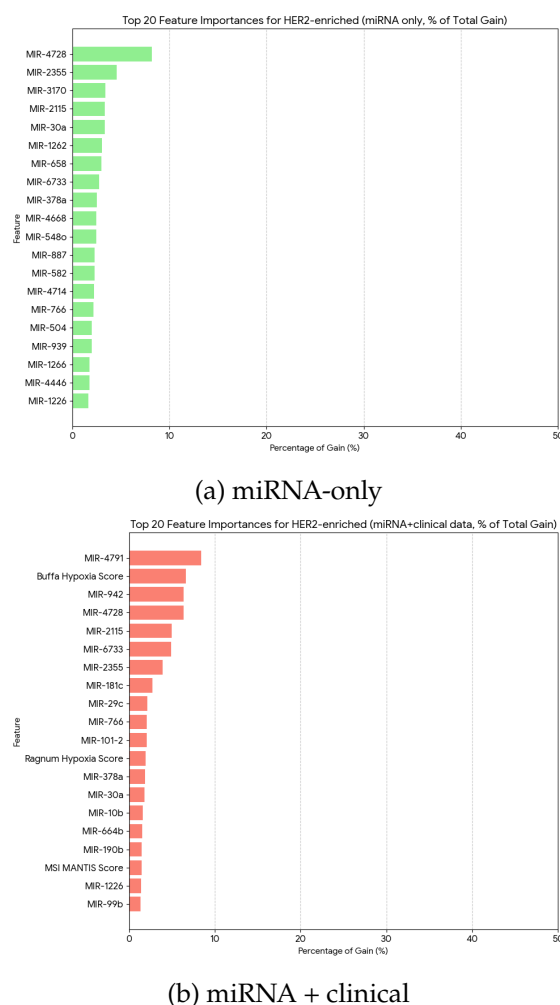
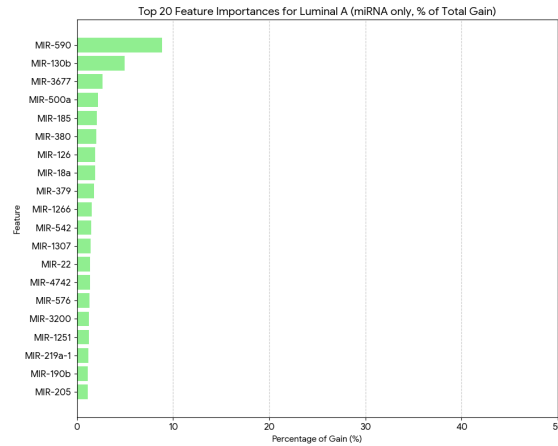


Figure 4.9: Top 20 XGBoost gain importances for HER2-enriched subtype in the expanded cohort, shown side by side. **Left:** miRNA-only feature configuration. **Right:** miRNA + clinical features configuration.

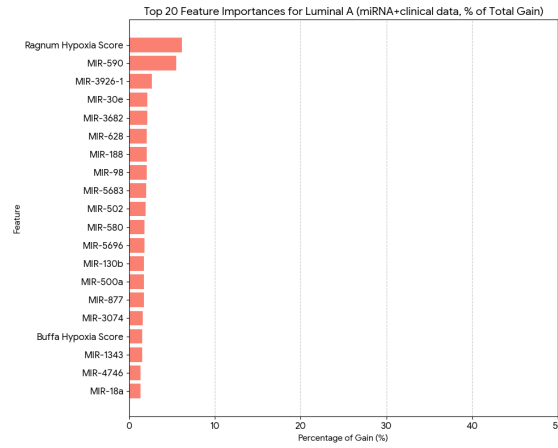
4.4.1.3 Luminal A Feature Panel

For Luminal A, the miRNA-only condition assigns the highest gain to **MIR-590** (8.84%), followed by MIR-130b (4.97%), with the remaining top-20 features each contributing less than 3%. The collective gain of the top-20 is 42.9%, indicating a moderate degree of concentration. In the miRNA + clinical condition, the **Ragnum Hypoxia Score** displaces MIR-590 as the leading feature (6.17% gain), with MIR-590 falling to second place (5.50%). The entry of a hypoxia score as the most informative feature in the multivariate context is a notable departure from the pilot study as this shift reflects the different clinical annotation available in the expanded cohort and illustrates how the identity of informative clinical features is *annotation-dependent* rather than intrinsic to the subtype biology.

The relatively diffuse importance structure for Luminal A (top-20 gain < 45% in both conditions) is consistent with this subtype’s large class size and its overlap with Luminal B: the model must leverage many features to partition the majority class.



(a) miRNA-only



(b) miRNA + clinical

Figure 4.10: Top 20 XGBoost gain importances for Luminal A subtype in the expanded cohort, shown side by side. **Left:** miRNA-only feature configuration. **Right:** miRNA + clinical features configuration.

4.4.1.4 Luminal B Feature Panel

The Luminal B importance landscape shows one notable shift relative to the pilot study. In the miRNA + clinical condition, **MIR-190b** emerges as the dominant feature (8.90% gain), a position it did not occupy in the pilot cohort. MIR-193b (4.20%) and MIR-301a (3.04%) follow in both configurations, providing some stability.

As in the pilot study, the collective gain of the Luminal B top-20 features ($\approx 43\%$ in miRNA-only; $\approx 46\%$ with clinical) remains the lowest among all subtypes, confirming the broader observation that Luminal B is the **least concentrated** in the miRNA feature importance space, a direct consequence of its transcriptomic heterogeneity.

Now that we have the feature importances for each subtype, we can move on to the ANOVA F-test to understand if these findings have any statistical meaning when we look at the lens of an approach that is model-agnostic.

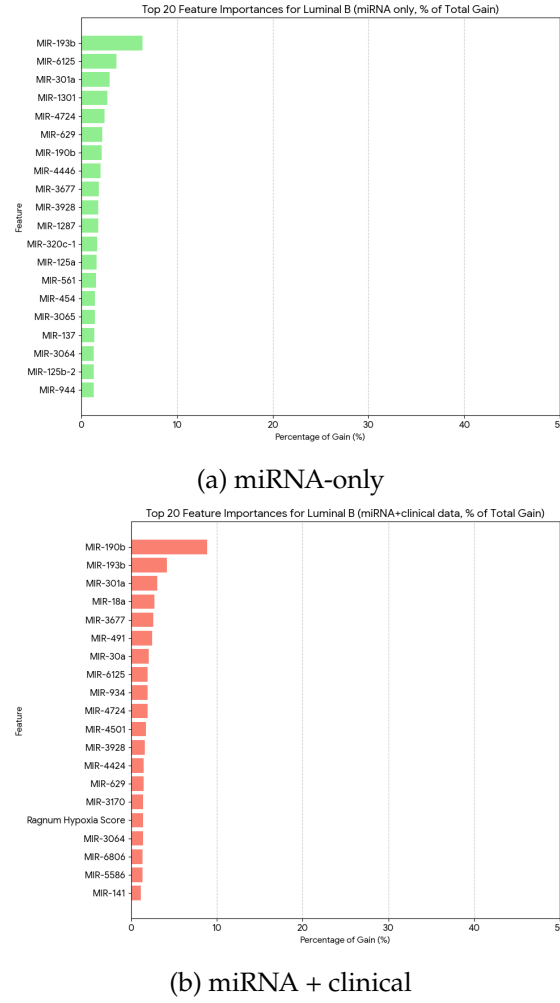


Figure 4.11: Top 20 XGBoost gain importances for Luminal B subtype in the expanded cohort, shown side by side. **Left:** miRNA-only feature configuration. **Right:** miRNA + clinical features configuration.

4.4.2 Statistical Corroboration via ANOVA F-Test

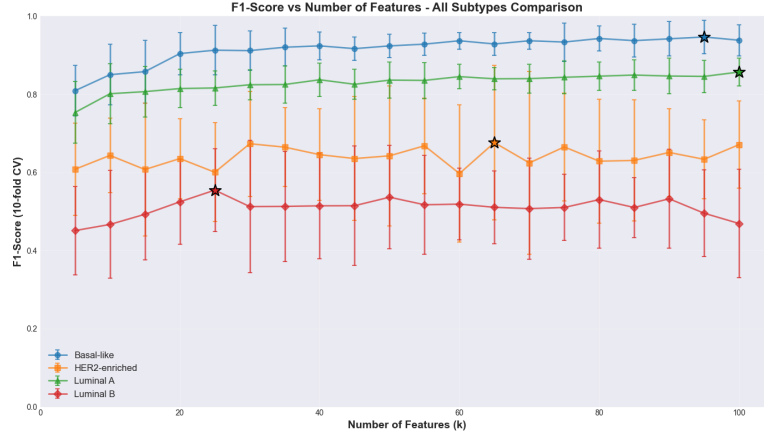
Following the two-level validation strategy introduced in Section 3.3.8, an ANOVA F-test pipeline was applied to the expanded cohort using the same implementation described in Section 3.3.8. For each subtype under OvR binarisation, features were ranked by their F-statistic and the top- k features were used to train a Random Forest classifier. F1-score was computed as a function of k (from $k = 5$ to $k = 100$) using 5-fold stratified cross-validation, with the same class-weighting and stratification strategy as in the pilot study. Both feature configurations (miRNA only; miRNA + clinical) were evaluated in parallel.

Table 4.9 summarises the key results from the ANOVA sweep. A comparison with the pilot-study values (Table 4.9) is possible at the level of best F1-score and initial F1-score (at $k = 5$), though the optimal k values are not directly comparable due to the different feature-space sizes.

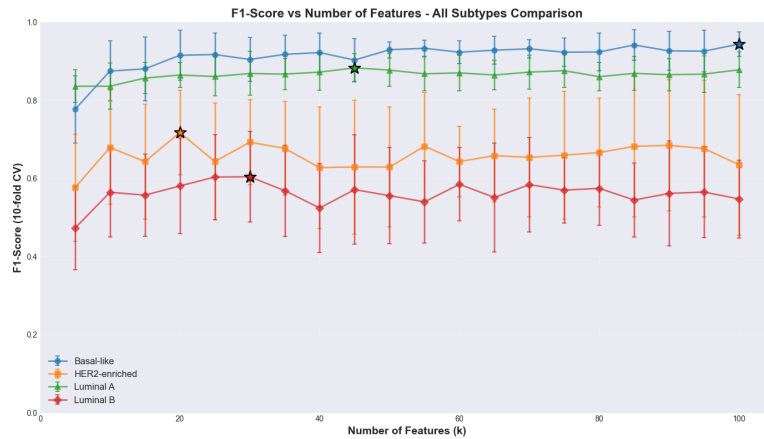
The ANOVA results replicate and strengthen several findings from the pilot study.

Table 4.9: ANOVA F-test pipeline summary for the expanded cohort ($n = 702$). For each subtype and feature configuration, the table reports the initial F1-score at $k = 5$ features, the best F1-score and the k at which it is achieved, and the improvement across the sweep.

Subtype	Condition	F1 at $k=5$	Best F1	Optimal k	Improvement
Basal-like	miRNA only	0.809	0.947	95	+0.138
	miRNA + clinical	0.777	0.944	100	+0.167
HER2-enriched	miRNA only	0.608	0.676	65	+0.068
	miRNA + clinical	0.576	0.718	20	+0.142
Luminal A	miRNA only	0.754	0.857	100	+0.104
	miRNA + clinical	0.836	0.883	45	+0.047
Luminal B	miRNA only	0.451	0.555	25	+0.104
	miRNA + clinical	0.473	0.684	30	+0.131



(a) with miRNA expression data only.



(b) with miRNA and clinical data.

Figure 4.12: Evolution of F1-score vs. k curves for all four subtypes under both conditions.

First, the **separability gradient across subtypes** is preserved: Basal-like achieves the highest best F1-score (0.947 / 0.944) regardless of condition, followed by Luminal A (0.857 /

0.883), HER2-enriched (0.676 / 0.718), and Luminal B (0.555 / 0.684). This ranking is consistent with both the XGBoost OvR results and the pilot-study ANOVA sweep, suggesting it reflects a genuine property of the data's underlying geometry rather than a model-specific or cohort-specific artifact.

Second, the **optimal feature count** for Basal-like (95–100 features) is substantially higher than for HER2-enriched (20–65) and Luminal B (25–30), despite Basal-like being the most separable subtype. This counterintuitive observation replicates the pilot study and can be understood as follows: Basal-like has a very strong signal that is distributed across many correlated miRNAs (as evidenced by the broad F-score distribution), whereas for HER2-enriched and Luminal B the incremental contribution of additional features is limited, and the performance plateaus or slightly degrades beyond a small number of top-ranked features.

Third, the addition of clinical features generally **improves** the ANOVA best F1-score for HER2-enriched (+0.042), Luminal A (+0.026) and Luminal B (+0.129), while leaving Basal-like virtually unchanged (−0.003). The large gain for Luminal B is particularly striking: with only 30 features, the combined model achieves an F1-score of 0.684 compared to 0.555 with miRNA alone. This suggests that at least some of the expanded cohort's genomic instability and hypoxia clinical features carry signals that are partially orthogonal to the miRNA expression space for this subtype.

4.4.2.1 Doubly-Validated miRNA Panels

As in the pilot study, the primary purpose of the ANOVA sweep is not to produce a standalone classifier, but to cross-reference its top-ranked features with those identified by the XGBoost booster, producing **doubly-validated candidate miRNA panels** that are supported by two independent methods: one multivariate and model-embedded, one marginal and model-agnostic. Features that survive both filters independently are the strongest biomarker candidates, because their discriminative power cannot be attributed to model-specific artefacts or to correlation with confounders.

The intersection of the ANOVA top- k list (at the subtype-specific optimal k) and the XGBoost top-20 gain ranking is presented in Figure 4.15 for all four subtypes and both feature configurations.

The doubly-validated panel for Basal-like is particularly robust: six miRNAs (MIR-577, MIR-934, MIR-18a, MIR-135b, MIR-190b, MIR-671) survive both filters under both feature configurations in the expanded cohort. Notably, the first three of these (MIR-577, MIR-934, MIR-135b) were also the highest-ranked doubly-validated features in the pilot study, representing a cross-cohort replication of the Basal-like miRNA panel. The consistency of these candidates across two independent datasets - different preprocessing, different clinical annotations, different cohort sizes - substantially increases our confidence in their role as biomarker candidates for Basal-like breast cancer.

For the remaining subtypes, the panel sizes are smaller, consistent with their lower

4.4. FEATURE EXTRACTION AND STATISTICAL CORROBORATION

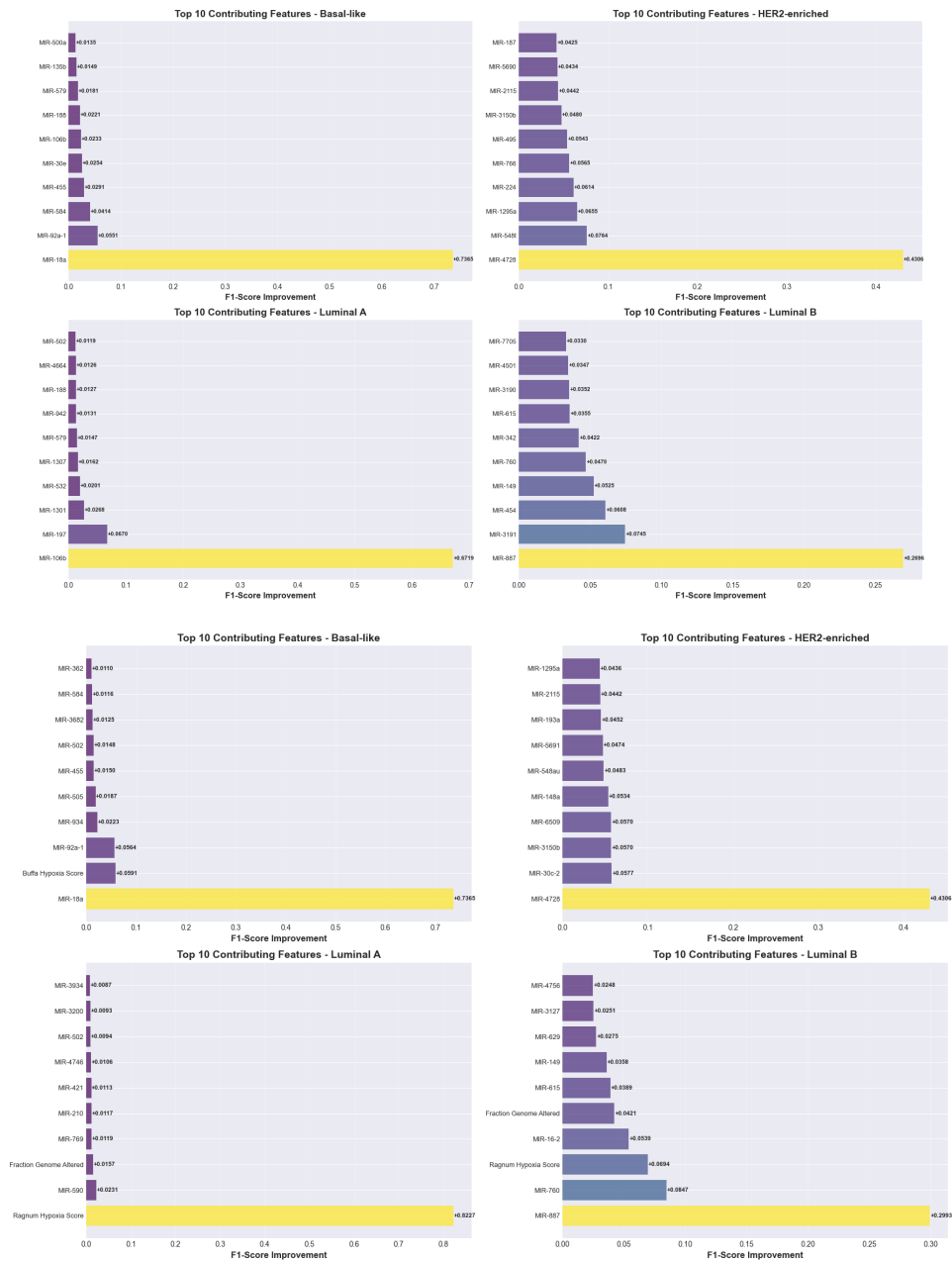
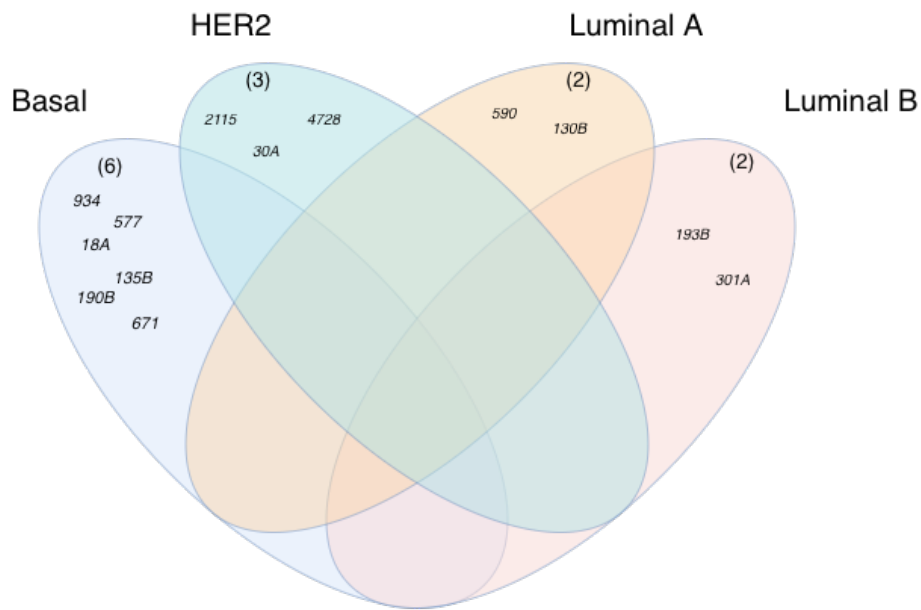
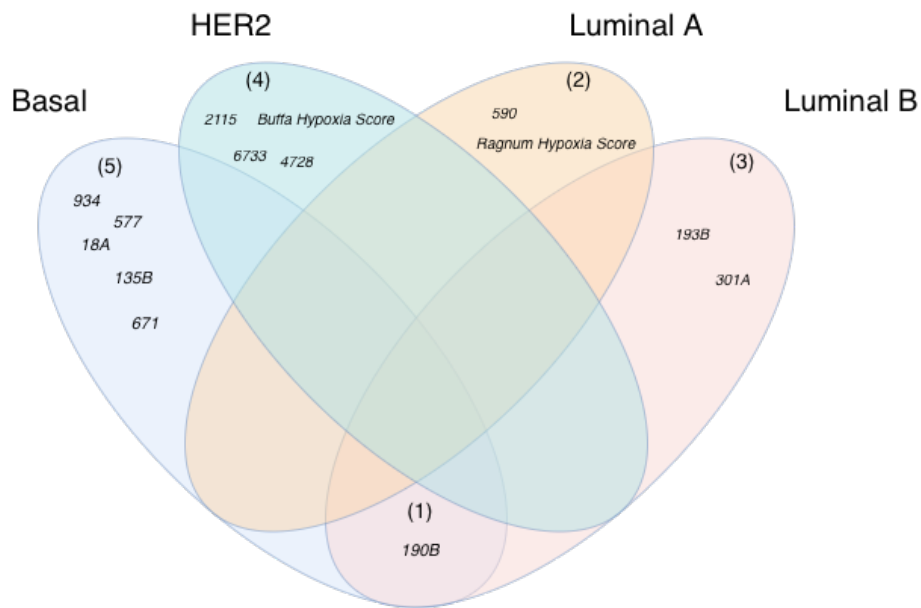


Figure 4.14: Top 10 ANOVA F-test features for Luminal A subtype under miRNA + clinical condition.

separability and more diffuse importance structures. The appearance of the Buffa Hypoxia Score for HER2-enriched and the Ragnum Hypoxia Score for Luminal A as doubly-validated clinical features is a dataset-specific finding that could not have been anticipated from the pilot study, which relied on receptor status variables. These results suggest that transcriptomic hypoxia signatures carry subtype-relevant biological information beyond what is captured by the miRNA expression data, at least within the expanded cohort's clinical annotation framework.



(a) miRNA only



(b) miRNA and Clinical data

Note: The prefix "MIR-" is omitted for brevity and to improve readability.

Figure 4.15: Feature intersections showing doubly-validated markers for all subtypes under both miRNA-only and miRNA+clinical conditions.

4.5 Group-Based Subtyping on the Expanded Cohort

In Chapter 3, the unsupervised clustering analysis (Section 3.4) established that K-Means, operating without any label information, recovers a data partition that aligns non-trivially with the known PAM50 subtype structure. The Basal-like subtype, in particular, was consistently isolated into a high-purity cluster in both the miRNA-only and miRNA + clinical conditions. The expanded cohort provides an opportunity to test whether this implicit structure in the miRNA expression space **persists and strengthens at scale**: with almost three times as many samples, the geometric separation between subtype-specific miRNA expression patterns should be more pronounced, provided that the patterns are genuine and not noise-driven.

The same K-Means pipeline described in Section 3.4.2 was applied without modification: z-score standardisation, $K = 4$ clusters (motivated in Section 3.4.2 by domain knowledge), K-Means++ initialisation and 10 random restarts. Both experimental conditions were evaluated: **miRNA only** (539 features) and **miRNA + clinical** (553 features). Labels were not used during clustering and were accessed only post-hoc to evaluate purity and cluster composition.

4.5.1 Cluster Quality Metrics

Table 4.10 summarises the aggregate evaluation metrics for both conditions on the expanded cohort, alongside the pilot-study values for direct comparison.

Table 4.10: K-Means performance summary for $K = 4$ on the expanded cohort ($n = 702$). Pilot-cohort values (from Table 3.14) are shown in grey for reference.

Condition	Cohort	Silhouette	Purity
miRNA only	Expanded ($n = 702$)	0.114	0.621
	Pilot ($n = 244$)	0.069	0.611
miRNA + clinical	Expanded ($n = 702$)	0.026	0.623
	Pilot ($n = 244$)	0.068	0.598

The aggregate purity in the expanded cohort is virtually identical to the pilot study under both conditions (0.621 vs. 0.611 for miRNA-only; 0.623 vs. 0.598 for miRNA + clinical), indicating that the degree of above-chance alignment between the discovered clusters and PAM50 subtypes is **stable across cohort sizes**. This stability is an important positive result: it suggests that the clustering signal is not an artefact of the pilot cohort's particular sample composition, but a consistent property of the TCGA-BRCA miRNA expression space.

The silhouette score, however, behaves differently across conditions. In the miRNA-only setting, it increases from 0.069 to 0.114 on the larger cohort, a relative improvement of 65%, indicating that the four-cluster partition is *geometrically more coherent* at scale, even if the overall purity is unchanged. This is consistent with what one would expect if the

Basal-like cluster (which dominates the purity score) becomes more compact relative to between-cluster distances as more samples fill in the expression landscape.

Conversely, in the miRNA + clinical condition, the silhouette score *drops* sharply from 0.068 to 0.026, while purity remains comparable (0.623). This apparent contradiction, near-identical purity but near-zero silhouette, is explained by the cluster structure detailed in the next section. The addition of 15 clinical features whose directions in feature space do not align with PAM50 subtype boundaries is sufficient to distort the Euclidean geometry used by K-Means, creating elongated or irregularly shaped clusters that are geometrically poorly separated but still dominated by a single subtype. In short, clinical features help maintain *label alignment* but severely disrupt *geometric compactness*, an observation that reinforces the pilot study’s conclusion that the subtype-discriminating signal resides primarily in the miRNA expression space.

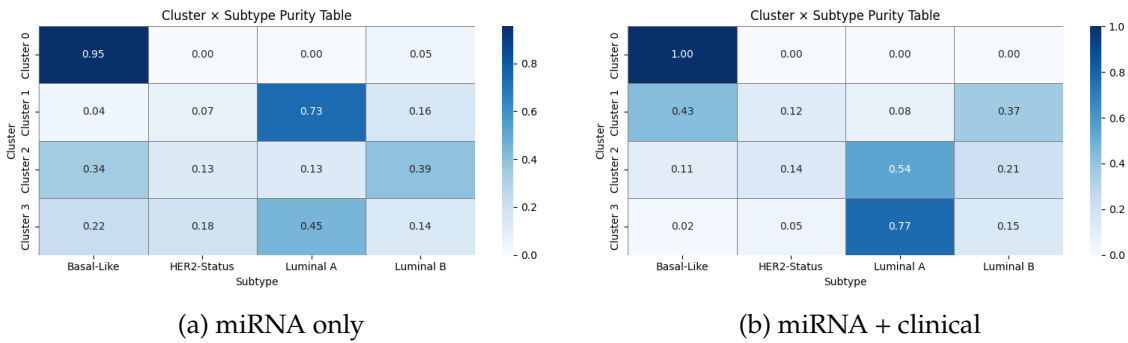


Figure 4.16: Cluster $\times$ Subtype purity heatmaps for both conditions. Each cell reports the proportion of samples of that subtype within the cluster. Note that cluster indices are not directly comparable across conditions; the same biological structure may appear under a different index in each panel.

4.5.2 Cluster Composition and Subtype Recovery

Aiming this approach to our main objective of finding miRNA biomarkers, we are interested in understanding the composition of the clusters that and map those evidences with the purity of each cluster to see if there is a correlation between the two and the findings of the supervised analysis. The per-cluster composition for both conditions is summarised in Table 4.11. As noted in Section 3.4.3, K-Means cluster indices are arbitrary and clusters are identified by their dominant subtype rather than by number.

Two findings from Table 4.11 warrant particular attention.

Basal-like isolation is preserved and strengthened. Under the miRNA-only condition, a small but extremely pure cluster (C0, $n = 22$, purity 95.5%) is dominated by Basal-like samples. Under the miRNA + clinical condition, this isolation becomes perfect: C0 ($n = 10$, purity 100%) contains exclusively Basal-like samples. The fact that K-Means, operating without labels, spontaneously isolates Basal-like tumours with near-perfect purity

Table 4.11: Cluster composition in the expanded cohort ($K = 4, n = 702$). Each cell reports the raw count of samples of that subtype. The dominant subtype per cluster is given in the rightmost column with its within-cluster purity.

Cond.	Cluster	Basal	HER2	Lum. A	Lum. B	Dominant (purity)
miRNA only	C0 ($n = 22$)	21	0	0	1	Basal-like (95.5%)
	C1 ($n = 428$)	15	32	313	68	Luminal A (73.1%)
	C2 ($n = 203$)	70	26	27	80	Luminal B (39.4%)
	C3 ($n = 49$)	11	9	22	7	Luminal A (44.9%)
miRNA +clin.	C0 ($n = 10$)	10	0	0	0	Basal-like (100.0%)
	C1 ($n = 186$)	80	23	15	68	Basal-like (43.0%)
	C2 ($n = 189$)	20	27	102	40	Luminal A (54.0%)
	C3 ($n = 317$)	7	17	245	48	Luminal A (77.3%)

on a 702-sample cohort constitutes strong evidence that the Basal-like miRNA expression signature occupies a **geometrically isolated region** of the expression space, a structural property rather than a classifier artefact.

Luminal A dominates the large clusters. In the miRNA-only condition, C1 ($n = 428$, purity 73.1%) is predominantly Luminal A, forming the largest and most coherent cluster after the Basal-like one. With clinical features, this structure is partially disrupted and redistributed across two large clusters (C2: 54% Luminal A, $n = 189$; C3: 77.3% Luminal A, $n = 317$), with C3 achieving a higher within-cluster purity. The emergence of two Luminal A-dominated clusters under the combined condition may reflect the known heterogeneity within Luminal A, as clinical features introduce additional axes of variation that K-Means exploits to further subdivide the majority class.

HER2-enriched and Luminal B are not reliably isolated. Neither HER2-enriched nor Luminal B forms a high-purity cluster in either condition. HER2-enriched samples are distributed across multiple clusters, and Luminal B achieves its highest within-cluster representation (39.4%) in a mixed cluster (C2) that also contains substantial Basal-like (34.5%) and HER2-enriched (12.8%) fractions. This lack of isolation is perfectly consistent with the supervised analysis: both subtypes are characterised by diffuse and overlapping miRNA expression patterns that do not map to geometrically compact regions in the standardised feature space. K-Means cannot recover what the supervised classifier also struggles to separate.

4.5.3 Cluster Feature Signatures

For each cluster, the miRNAs that most strongly characterise its centroid relative to the rest of the data were extracted using the absolute Z-score importance method described in Section 3.4.3.3. The top-20 features for each cluster and both conditions are presented

below. The interpretation focuses on the two clusters with clear subtype identity (Basal-like and the dominant Luminal A cluster), as these are the only cases where a biological interpretation of the centroid is meaningful.

Basal-like cluster (miRNA only, C0). The top features for the Basal-like cluster (Figure 4.19a) include MIR-590, MIR-18a, MIR-577, MIR-130b, MIR-576 and MIR-197. The overlap between this unsupervised centroid signature and the XGBoost gain importances for Basal-like, in particular the consistent appearance of MIR-577 and MIR-18a across both methods, provides cross-paradigm corroboration for these miRNAs as Basal-like discriminators.

Basal-like cluster (miRNA + clinical, C0). With clinical features (Figure 4.20a), the Basal-like centroid signature is largely consistent with the miRNA-only case: MIR-577, MIR-576, MIR-18a, MIR-590, MIR-4766 and MIR-130b dominate. The increased Z-score magnitudes suggest that the Basal-like centroid is even more clearly separated from the rest of the data in the combined feature space, consistent with the 100% purity of this cluster.

Dominant Luminal A cluster (miRNA only, C1). The Luminal A-dominated cluster C1 shows a much less concentrated signature. The top features have markedly lower Z-scores (≤ 0.47), reflecting the absence of a small set of dominant Luminal A-specific miRNAs. The leading features include MIR-197, MIR-324, MIR-1d, MIR-1306, and MIR-33a - a distinct set from the XGBoost gain importances for Luminal A, suggesting that different aspects of the Luminal A miRNA expression landscape are captured by the centroid geometry versus the multivariate decision boundary.

miRNA Only: Cluster Distribution



Figure 4.17: Breast Cancer Subtype Distribution: miRNA Only Dataset.

miRNA + Clinical Data: Cluster Distribution

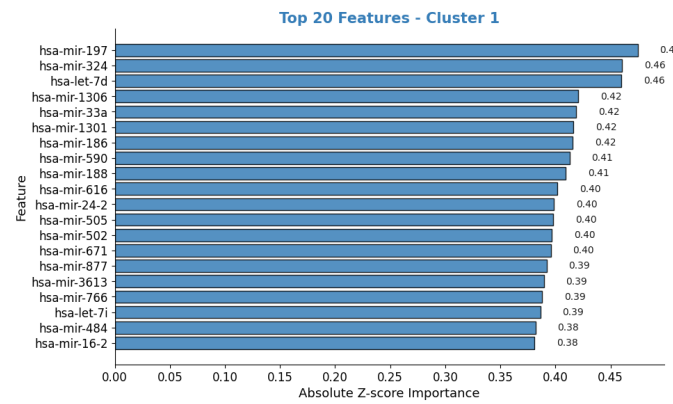


Figure 4.18: Breast Cancer Subtype Distribution: miRNA + Clinical Dataset.

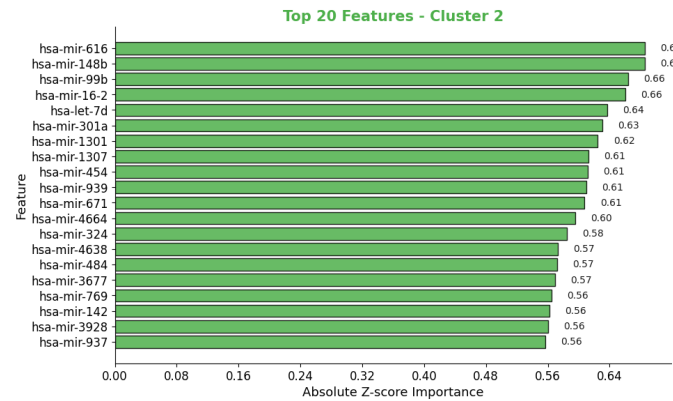
4.5. GROUP-BASED SUBTYPING ON THE EXPANDED COHORT



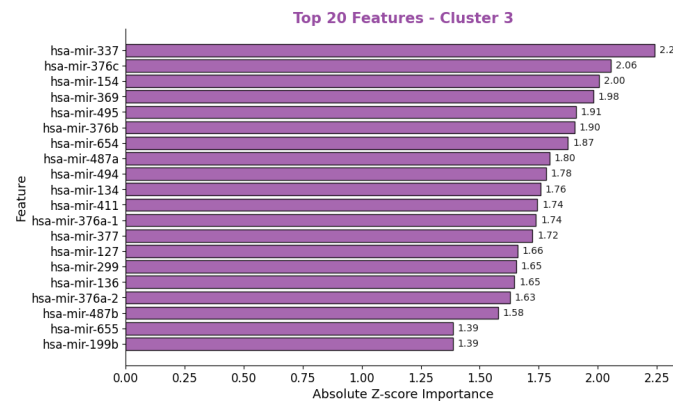
(a) Cluster 0



(b) Cluster 1



(c) Cluster 2



(d) Cluster 3

Figure 4.19: Top-20 absolute centroid z-score feature rankings for each K-Means cluster with miRNA-only features, where each panel shows the miRNAs most strongly characteristic of the cluster. (Cluster assignments and correspondence are described in Table 3.15.)

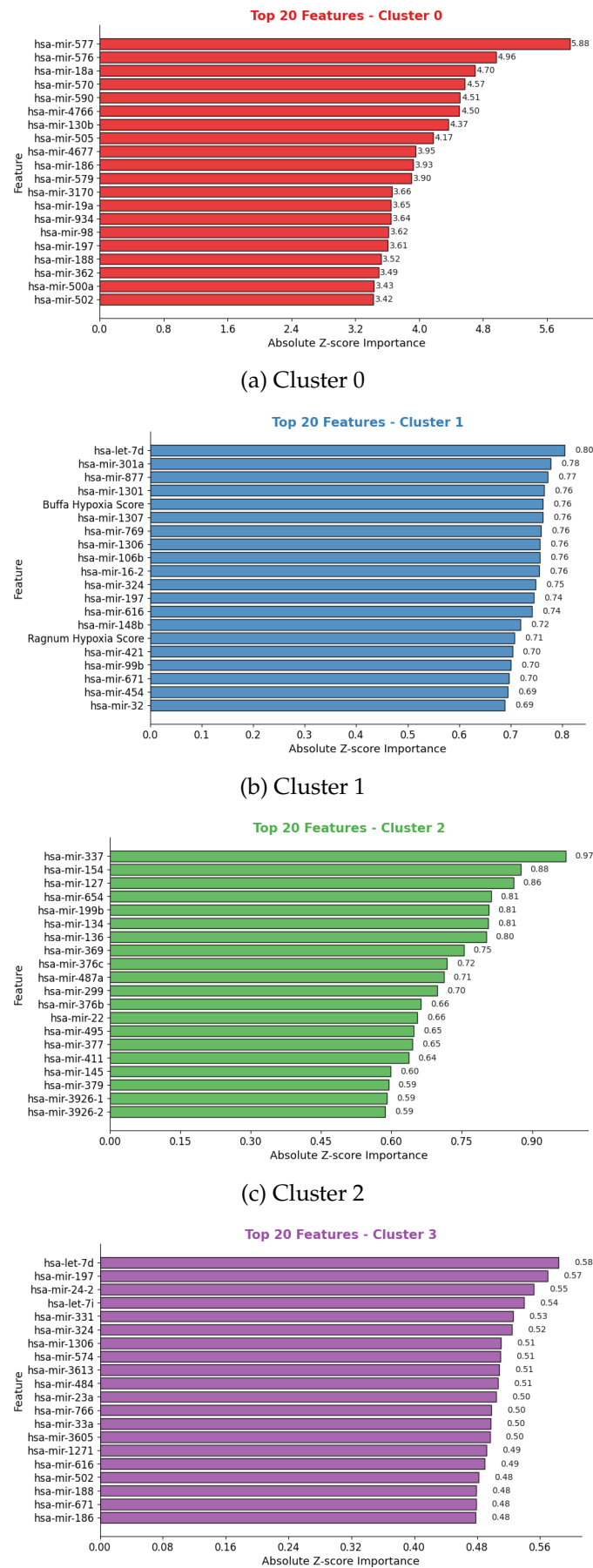


Figure 4.20: Top-20 absolute centroid z-score feature rankings for each K-Means cluster with miRNA and clinical features, where each panel shows the features that most strongly characterize the cluster (Cluster assignments and correspondence are described in Table 4.17).

4.5.4 Summary

The K-Means analysis on the expanded cohort confirms and extends the pilot study's findings on three key points. First, the **Basal-like subtype occupies a geometrically isolated region** of the miRNA expression space: K-Means recovers a near-pure Basal-like cluster without any label supervision, and this isolation becomes perfect when clinical features are added. Second, the **clustering purity is stable across cohort sizes**: expanding from 244 to 702 samples does not meaningfully change the proportion of correctly isolated samples, suggesting the result is not cohort-size-dependent. Third, **HER2-enriched and Luminal B remain inseparable by unsupervised means**: neither subtype forms a high-purity cluster under any condition, directly mirroring their difficulty in the supervised setting and reinforcing the conclusion that their miRNA signatures lack the geometric compactness required for cluster recovery.

The convergence between the supervised (XGBoost, ANOVA) and unsupervised (K-Means) perspectives - centred on the same core miRNAs (particularly MIR-577 and MIR-18a) for the Basal-like subtype, and on the same separability gradient across subtypes - is the strongest methodological finding of this chapter, as it is fully independent of any labelled supervision. The detailed cross-cohort comparison of feature panels is reserved for Chapter 5, where the results of both the pilot and expanded analyses are synthesised in the light of the broader literature.

4.6 Supervised and Unsupervised Perspectives: A Cross-Paradigm Comparison

The preceding sections approached the expanded cohort from two fundamentally different angles. The supervised analysis asked: *given that we know the subtype labels, which features does a model need to classify each subtype correctly, and are those features statistically meaningful in isolation?* The unsupervised analysis asked instead: *ignoring the labels entirely, does the miRNA expression data spontaneously organise into groups that correspond to the known biology?* Because the two paradigms make no shared assumptions whatsoever, any finding that emerges consistently from both carries an evidential weight that neither method alone could establish. This section brings those two perspectives together, examining where they agree, where they diverge, and what those convergences and divergences reveal about each subtype.

4.6.1 Basal-Like: Convergent Evidence from Both Paradigms

Basal-like is the subtype around which the supervised and unsupervised results converge most completely. On the supervised side, the OvR XGBoost classifier achieved a near-perfect F1-score of 0.98 in the miRNA-only condition and a perfect 1.00 with clinical

features, with MIR-577 alone accounting for 46.6% and 64.2% of the total booster gain respectively. The ANOVA sweep independently confirmed that Basal-like features carry the highest marginal statistical separation across all subtypes, with the best cross-validated F1-score reaching 0.947. On the unsupervised side, K-Means isolated Basal-like into a cluster of 95.5% purity (miRNA-only) that became perfectly pure (100%) when clinical features were added, without ever observing a single label.

Beyond these aggregate metrics, the feature-level agreement is particularly striking. MIR-577 and MIR-18a appear as top-ranked features in the XGBoost booster importances, in the ANOVA top-20 list and in the centroid Z-score signature of the Basal-like K-Means cluster, under both feature configurations and across both cohorts. This triple-method, dual-cohort replication places these two miRNAs in the strongest possible position as biomarker candidates: they are simultaneously important to a multivariate classifier, statistically separating in isolation, and geometrically defining of the Basal-like cluster in expression space. The conclusion that emerges from both paradigms is the same and reinforces each other: **Basal-like tumours occupy a distinct, compact, and well-defined region of the miRNA expression landscape**, and that region can be navigated by both supervised and unsupervised means.

4.6.2 Luminal A: Consistent Performance, Inconsistent Geometry

Luminal A presents a more nuanced picture. The supervised analysis classified it reliably, macro-averaged F1-score of 0.89 in the miRNA-only condition, recall of 0.96 and identified MIR-590 as the leading feature by XGBoost gain (Figure 4.10), confirmed by the ANOVA sweep. These results suggest a learnable signal. However, the unsupervised analysis complicates this reading: K-Means never isolates Luminal A into a single high-purity cluster. In the miRNA-only condition it is spread across two clusters (C1: 73.1% purity, $n = 428$; C3: 44.9% purity, $n = 49$) (Figure 4.16), and with clinical features it splits further into two large, Luminal A-dominated clusters of different purities (C3: 77.3%, $n = 317$; C2: 54.0%, $n = 189$).

The apparent contradiction, namely reliable supervised classification but no clean unsupervised isolation, is actually informative. It reflects the fundamental difference between what the two methods are measuring. The OvR classifier learns a *decision boundary* that separates Luminal A from the union of all other subtypes; it does not require Luminal A to be internally homogeneous, only that the boundary can be drawn. K-Means, by contrast, requires Luminal A to form a *compact, convex region* in the standardised feature space, a considerably stronger geometric demand. The fact that K-Means repeatedly splits Luminal A into two sub-clusters, rather than conflating it with another subtype, suggests that the subtype harbours genuine internal heterogeneity: there may be two biologically distinct sub-populations within the Luminal A label that the PAM50 classification does not resolve. The K-Means centroid features for these two Luminal A-dominated clusters diverge somewhat from those identified by the supervised analysis, reinforcing the idea

that the unsupervised partition is capturing a different dimension of the data's structure.

4.6.3 HER2-Enriched and Luminal B: Consistent Difficulty Across Both Paradigms

HER2-enriched and Luminal B are the two subtypes where the supervised and unsupervised perspectives tell the same negative story. From the supervised side, the OvR classifiers for both subtypes achieved substantially lower F1-scores than Basal-like and Luminal A, 0.42 and 0.56 respectively in the miRNA-only condition (Tables 4.6 and 4.8), despite access to labelled training data and a model explicitly optimised to find discriminative boundaries. The ANOVA sweep produced the lowest best F1-scores for these two subtypes (0.676 for HER2-enriched; 0.555 for Luminal B), and the feature importance structures were diffuse: the top-20 XGBoost features accounted for only 57.1% and 43.0% of the total booster gain respectively in the miRNA-only condition, compared to 92.0% for Basal-like.

From the unsupervised side, neither subtype formed a high-purity cluster in any condition. HER2-enriched samples were distributed across all four clusters in both conditions, and the best within-cluster Luminal B representation reached only 39.4% (Figure 4.16) in a cluster also heavily populated by Basal-like (34.5%) and HER2-enriched (12.8%) samples. K-Means, operating without labels, cannot find what the supervised classifier (operating with labels) also struggles to find.

The alignment between the two paradigms here is not a failure but a finding. It rules out the possibility that the supervised classifier's limited performance is a model-specific limitation that a different algorithm might overcome: if the signal were there in the miRNA expression geometry, K-Means would detect it as a compact cluster, as it does for Basal-like. The convergent evidence instead points to a data-intrinsic property, the miRNA expression profiles of HER2-enriched and Luminal B **do not occupy geometrically separable regions** of the expression space, meaning that neither supervised nor unsupervised approaches can cleanly recover them from miRNA data alone. For HER2-enriched, the addition of clinical features (specifically the Buffa Hypoxia Score) provides a partial improvement in supervised classification, suggesting that information orthogonal to the miRNA space exists but is not sufficient to resolve the problem fully. For Luminal B, the primary limitation is its transcriptomic proximity to Luminal A, a proximity that both the supervised decision boundary and the unsupervised cluster geometry reflect consistently.

4.6.4 The Role of Clinical Features: A Divergent Effect Across Paradigms

One of the more nuanced findings to emerge from the cross-paradigm comparison concerns the effect of adding clinical features. In the supervised setting, clinical covariates improve classification for HER2-enriched (+0.15 F1) and Luminal B (+0.09 F1), with negligible change for Basal-like and Luminal A. In the ANOVA sweep, the clinical features

yield meaningful F1 gains for HER2-enriched (+0.042) and Luminal B (+0.129), again leaving Basal-like essentially unchanged.

In the unsupervised setting, the effect of clinical features is quite different. Overall purity changes little (+0.002 from 0.621 to 0.623), but the silhouette score collapses from 0.114 to 0.026, a near-fourfold drop in geometric coherence. The structural consequence is visible in the cluster composition: the addition of clinical features converts the clean Luminal A majority cluster seen under miRNA-only conditions into two large, overlapping Luminal A-dominated clusters, while the Basal-like cluster shrinks from 22 samples to 10 but becomes perfectly pure. The clinical features are effectively introducing axes of variation that K-Means exploits to subdivide the data, but those axes do not align with subtype boundaries - they pull the clusters apart geometrically without improving their biological coherence.

The contrast is sharp and interpretable: **clinical features carry discriminative signal that a supervised model can leverage against a labelled boundary, but they introduce geometric noise that disrupts the unsupervised structure.** This asymmetry is a direct consequence of how the two methods use information. A supervised classifier learns *which* dimensions matter for each label; an unsupervised algorithm treats all dimensions symmetrically. In a feature space where miRNA dimensions encode subtype structure and clinical dimensions encode orthogonal biological variation, the supervised approach correctly weights the informative dimensions while K-Means is pulled in multiple directions at once. This distinction has a practical implication for biomarker discovery: clinical features are more valuable as supervised covariates than as clustering inputs, and the miRNA expression space is the more reliable substrate for unsupervised subtype recovery.